extranuclear, planetary electron is detached and hurled out of the atom, or that the extranuclear structure itself remains in an excited state, from which it will revert to its original state through the emission of a photon only. But whereas the electron is ejected in the very moment of the interference, the state in which the photon is formed and emitted does not ensue immediately; a certain length of time must always elapse first. This length of time is very short in absolute duration, but of considerable length relative to the time required for a complete revolution of a planetary electron around the nucleus. We may regard this as a characteristic which distinguishes true elementary particles from secondarily formed ones.

Proceeding from the same viewpoint, let us now examine those elementary particles which merit immediate consideration as building blocks of the nucleus. The nucleus, too, can be subjected to outside interference, in the form of bombardment by alpha particles or other elementary particles. As we have already seen, nuclear reactions may then take place, in which either a proton or a neutron is hurled forth. There are also instances where alpha particles are ejected. But it is certain that the latter are not true elementary particles. The ejection of a proton or neutron—like that of an extranuclear, planetary electron—usually occurs in the moment of interference. But in such a nuclear reaction an unstable, radioactive atom may be produced, which then changes further by a radioactive process. In these cases, only electrons or positrons, as well as neutrinos, are emitted. Like the naturally radioactive substances, these radioactive atoms have a definite decay probability or average life, which varies according to the individual instance. Thus, a longer or shorter period of time must elapse before the electron or positron is ejected, accompanied by a neutrino, as in the case of the emission of photons from the extranuclear atomic structure. On the other hand, there are also cases in which outside interference will result in the emission of a gamma-ray photon. Generally speaking, the time interval is long when measured in terms of nuclear frequencies, but extremely short in absolute duration, and it cannot be measured directly, but only by inference from other data. It occasionally happens, of course, that a gamma-ray photon is ejected in the very moment of the interference.

From the above findings we can conclude that only the protons and the neutrons may be regarded as true building blocks of the nucleus. This conclusion is very close to Prout’s old hypothesis. The mass of a neutron differs very little from that of a proton.

The ideas discussed are illustrated in the following table by a comparative survey of the conditions which obtain in the extranuclear structure of the atom, on the one hand, and those present in the nucleus, on the other hand.